

2. AI-Powered Code Explainer

Objective:

Build a web-based or CLI tool that takes a code snippet as input and outputs a plain-English explanation of what the code does.

Requirements (MVP):

- Accept code in Python or JavaScript as input from a user
- Use an AI model to generate a plain-English explanation (~2–4 sentences)
- Highlight key parts of the code (e.g., functions, logic blocks) if using a GUI
- Allow the user to submit multiple snippets and view each explanation

Constraints:

- Use any public LLM (OpenAI, Claude, Mistral, etc.)
- Bonus: Use syntax parsing (e.g., AST) to annotate code before sending to LLM
- Must complete within 2 business days
- Be ready to discuss how you handle hallucinations and code accuracy

Bonus:

- Add diff-view comparison between the original code and an "optimized" version suggested by the AI
- Explain time/space complexity if detectable

Submission:

Please submit within 1-2 business days:

- A GitHub repo containing your working code
- A summary in the README describing:
 - Your system architecture and technical decisions
 - The AI tool(s) you selected and your reasoning

A member of our team will review your submission. If selected, be ready to present your solution to a senior engineer in a live technical discussion.